

Total No. of Questions : 8]

SEAT No. :

PD4250

[Total No. of Pages : 2

[6403]-45

T.E. (Computer Engineering)
ARTIFICIAL INTELLIGENCE
(2019 Pattern) (Semester - VI) (310253)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Assume Suitable data if necessary.*

- Q1)** a) Explain Alpha-Beta tree search and cut off procedure with example. [9]
b) Explain in details the concept of backtracking and constraint propagation to solve n-queens problem. [8]

OR

- Q2)** a) What is constraint satisfaction problem? Explain with example. [5]
b) Compare and contrast the stochastic games and partial observable games. [4]
c) How AI technique is used to solve tic-tac-toe problem. [8]

- Q3)** a) Explain Wumpus world environment with PEAS description. [9]
b) What is knowledge representation in propositional logic? Compare and contrast PL and FOL. [8]

OR

- Q4)** a) Explain different inference rules in FOL with suitable example. [9]
b) What is an agent? Explain knowledge based agent with architecture diagram, also state the significance of inference engine. [8]

- Q5)** a) Illustrate with an example the use of the unification algorithm to prove the concept of resolution. [9]
b) Define and explain the forward changing with example, analyze the differences between forward and backward changing. [9]

OR

P.T.O.

Q6) a) Describe Ontological Engineering w.r.t. Categories, Objects and Model. [9]

b) Define First order Logic, Explain FOL inference for following classifiers. [9]

i) Universal Generalization

ii) Universal Instantiation

iii) Extential Instantiation

iv) Extential Introduction

Q7) a) Explain the algorithm for classical planning with an example. [6]

b) Analyze various planning approaches in AI. [6]

c) Explain the Hierarchical planning with relevant example. [6]

OR

Q8) a) Explain with example how planning is different than problem solving. [6]

b) Explain types of planning in details. [6]

c) Explain AI components and AI Architecture. [6]

